178010: lithic–quartz sandstone, Mount Olive

Location and sampling

NULLAGINE (SF 51-5), EASTERN CREEK (3054)
MGA Zone 51, 250130E 7615210N

Sampled on 25 May 2002

The sample was taken from a 1.5 m-diameter block located 3 m west of the access track, 600 m west of Mount Olive, on a rocky ridge at least 30 m high and at the northern entry to a rocky gorge.

Tectonic unit/relations

The sample is from a dark-grey, poorly sorted, coarse-grained sandstone near the base of the Mosquito Creek Formation, East Pilbara Granite–Greenstone Terrane (Farrell, in prep.). The sandstone was collected to constrain the depositional age of the basal part of the Mosquito Creek Formation, and obtain provenance data.

Petrographic description

This dark-grey, weakly foliated rock contains abundant (35 vol.%) single-crystal quartz grains, from 0.2 to 1 mm in diameter, accompanied by chert clasts up to 1.5 mm in diameter (45 vol.%). It is a coarse-grained lithic (chert) – quartz sandstone, with minor muscovite and carbonaceous clasts, and transected by parallel limonite-filled fractures. Crystals altered by pyrite and hematite or goethite are disseminated, with rare carbonate in some of the chert clasts. A few chert clasts contain disseminated carbonate as small rhombs, but others have limonite or sericite, passing into clasts composed of decussate fine-grained sericite. The larger clasts have a subparallel orientation apparently parallel to bedding. There are also massive and laminated quartz-rich clasts that could represent silicified acid volcanic fragments. Some of the large single-crystal quartz grains have inclusions of apatite, suggesting an igneous provenance, but others have inclusions of vermiform chlorite, indicating derivation from vein-quartz. Rare detrital muscovite forms flakes, mostly less than 1 mm long, and rarely up to 2 mm long, and commonly rimmed by limonite. Flakes altered to ?kaolin and diffuse leucoxene could have been biotite. There are small clasts rich in carbonaceous matter, but they are not abundant enough to account for the overall grey colour of the hand specimen. Other clasts contain leucoxene with or without quartz, and there are clasts of chlorite, sericite, or illite and quartz. There is also rare pyrite, about 0.2 mm in grain size, and more abundant patches of limonite, hematite, goethite, or hematite–goethite, apparently replacing euhedral crystals of ?pyrite. Some crystals are coated by flakes of muscovite. Stylolite-like veins are abundant parallel to the layering and are filled with goethite. The metamorphism of this sample seems to have been weak and at low to very low grade.

Zircon morphology

The zircons isolated from this sample have a range of morphologies, but are typically colourless, pale yellowish-brown or dark brown, between 90 × 140 µm and 160 × 280 µm in size, and equant or slightly elongate, and subhedral. Many grains are fractured and many show evidence of surface pitting or abrasion, consistent with detrital transport. Fluid and mineral inclusions are common. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 30 January and 6 February 2003. The counter deadtime was 24 ns during both analysis sessions. During the first analysis sesssion, nine analyses of the CZ3 standard obtained during the analysis session indicated a Pb*/U calibration uncertainty of 1.40% (1σ). Analyses 1.1 to 18.1 were obtained during the first analysis session. During the second analysis session, five analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 4.18% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 4.1, 7.1, and 18.1, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Twenty-seven analyses were obtained from 27 zircons. Results are given in Table 1, and shown on concordia and Gaussian-summation probability density plots in Figures 2 and 3 respectively.

Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with at least one episode of radiogenic-Pb loss. On the basis of their $^{207}Pb/^{206}Pb$ ratios, many analyses can be assigned to one of three groups. Concordant analyses 9.1, 21.1, and 27.1, and slightly discordant analysis 24.1, assigned to Group 1, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3294 ± 14 Ma (chi-squared = 0.52). Concordant analyses 21.1 and 12.1, and slightly discordant analyses 5.1 and 8.1, assigned to Group 2, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3424 ± 12 Ma (chi-squared = 0.35). Eleven concordant and slightly discordant analyses of 11 zircons (3.1, 6.1, 7.1, 10.1, 11.1, 13.1, 16.1, 18.1, 19.1, 20.1, 22.1), assigned to Group 3, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3461 ± 7 Ma (chi-squared = 1.50). The remaining analyses cannot be confidently grouped.

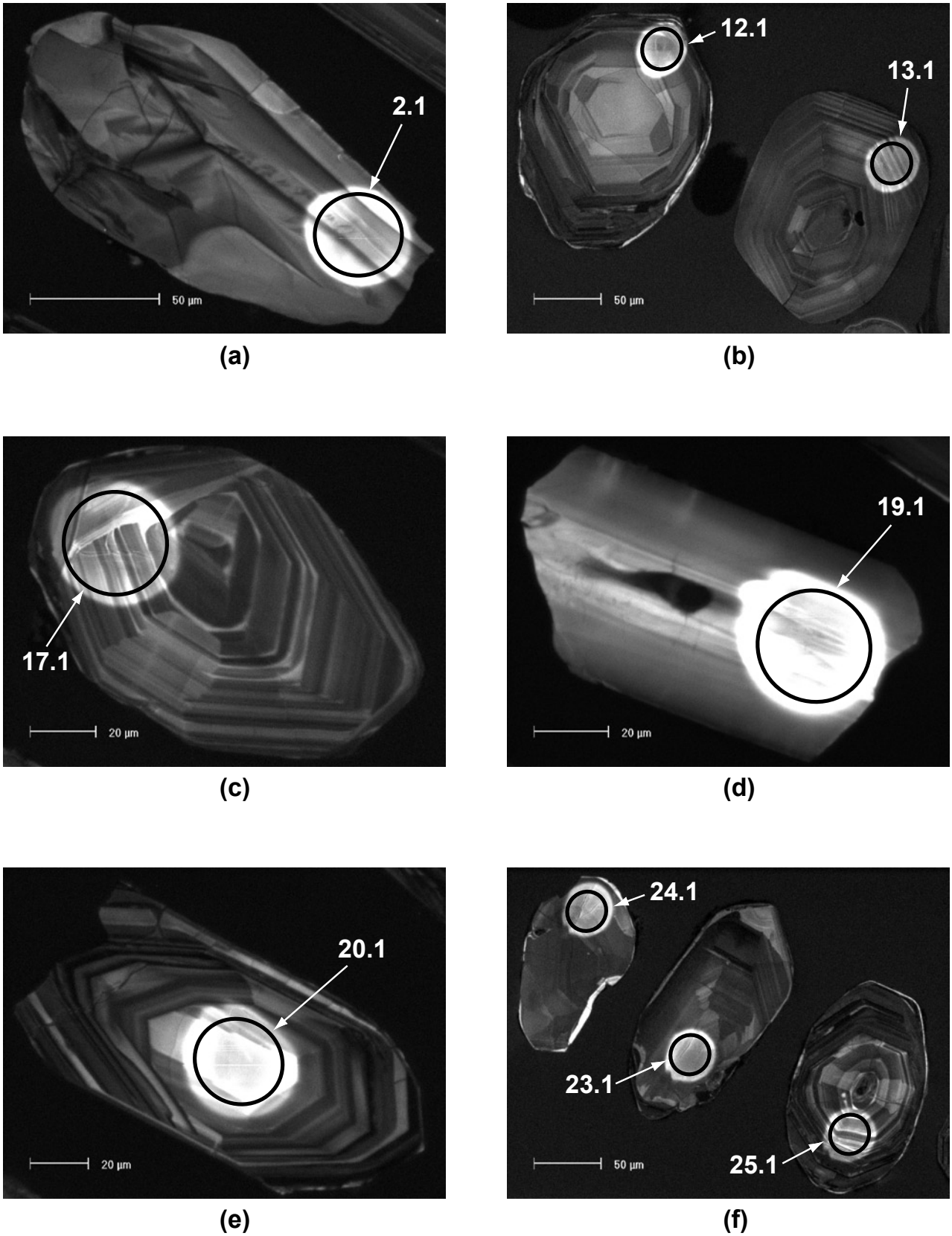

Figure 1. **Cathodoluminescence images of representative zircons from sample 178010: lithic–quartz sandstone, Mount Olive**

Table 1. Ion microprobe analytical results for zircons from sample 178010: lithic–quartz sandstone, Mount Olive

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	$\pm 1\sigma$	$^{208}Pb/^{206}Pb$	$\pm 1\sigma$	$^{206}Pb/^{238}U$	$\pm 1\sigma$	$^{207}Pb/^{235}U$	$\pm 1\sigma$	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	$\pm 1\sigma$
1.1	97	148	97	0.073	0.27235	0.00151	0.39810	0.00261	0.6921	0.0111	25.990	0.456	102	3 320	9
2.1	61	30	55	0.085	0.29252	0.00206	0.12589	0.00268	0.7326	0.0124	29.548	0.565	103	3 431	11
3.1	140	157	140	0.070	0.29671	0.00122	0.29129	0.00164	0.7303	0.0112	29.879	0.490	102	3 453	6
4.1	133	80	107	1.745	0.31543	0.00207	0.16167	0.00365	0.5962	0.0092	25.930	0.455	85	3 548	10
5.1	188	113	159	0.297	0.29027	0.00116	0.14588	0.00145	0.6808	0.0103	27.248	0.438	98	3 419	6
6.1	164	62	144	0.041	0.29964	0.00117	0.10086	0.00108	0.7284	0.0110	30.094	0.485	102	3 468	6
7.1	271	231	191	4.028	0.30017	0.00216	0.26075	0.00438	0.4637	0.0068	19.192	0.330	71	3 471	11
8.1	124	169	100	0.442	0.29192	0.00156	0.13156	0.00217	0.6490	0.0101	26.123	0.446	94	3 428	8
9.1	114	163	109	0.095	0.26705	0.00139	0.37475	0.00233	0.6777	0.0106	24.955	0.425	101	3 289	8
10.1	233	52	199	0.039	0.29797	0.00096	0.05629	0.00072	0.7306	0.0108	30.017	0.467	102	3 460	5
11.1	53	31	48	0.233	0.29568	0.00213	0.15120	0.00257	0.7243	0.0127	29.530	0.582	102	3 448	11
12.1	117	60	105	0.099	0.29149	0.00136	0.13415	0.00135	0.7308	0.0113	29.370	0.492	103	3 426	7
13.1	172	113	158	0.273	0.29959	0.00121	0.17325	0.00153	0.7189	0.0108	29.695	0.478	101	3 468	6
14.1	147	104	145	0.095	0.32574	0.00129	0.18334	0.00137	0.7572	0.0116	34.008	0.554	101	3 597	6
15.1	168	85	168	0.030	0.35156	0.00121	0.13282	0.00102	0.7841	0.0119	38.008	0.605	100	3 714	5
16.1	123	127	120	0.103	0.29699	0.00139	0.26851	0.00194	0.7253	0.0114	29.702	0.502	102	3 455	7
17.1	99	39	89	0.387	0.30407	0.00169	0.10149	0.00221	0.7311	0.0117	30.651	0.539	101	3 491	9
18.1	138	151	128	0.850	0.29641	0.00158	0.28454	0.00266	0.6648	0.0103	27.169	0.462	95	3 452	8
19.1	16	7	14	1.472	0.28951	0.00626	0.10108	0.01112	0.7159	0.0317	28.576	1.475	102	3 415	34
20.1	33	16	28	0.375	0.30263	0.00320	0.12105	0.00401	0.6857	0.0294	28.613	1.302	97	3 484	16
21.1	145	99	128	0.077	0.26759	0.00124	0.17423	0.00137	0.7086	0.0298	26.144	1.120	105	3 292	7
22.1	98	46	88	0.931	0.30204	0.00203	0.11835	0.00308	0.7168	0.0302	29.853	1.299	100	3 481	10
23.1	164	127	166	0.104	0.32069	0.00127	0.19670	0.00139	0.7706	0.0324	34.075	1.454	103	3 573	6
24.1	142	244	132	0.668	0.26848	0.00160	0.46849	0.00320	0.6114	0.0257	22.633	0.976	93	3 297	9
25.1	193	123	169	0.296	0.28638	0.00125	0.15781	0.00164	0.6960	0.0292	27.481	1.174	100	3 398	7
26.1	79	35	72	0.204	0.30562	0.00203	0.10383	0.00242	0.7458	0.0315	31.426	1.369	103	3 499	10
27.1	74	129	78	0.234	0.27010	0.00193	0.45910	0.00365	0.7018	0.0297	26.134	1.143	104	3 307	11

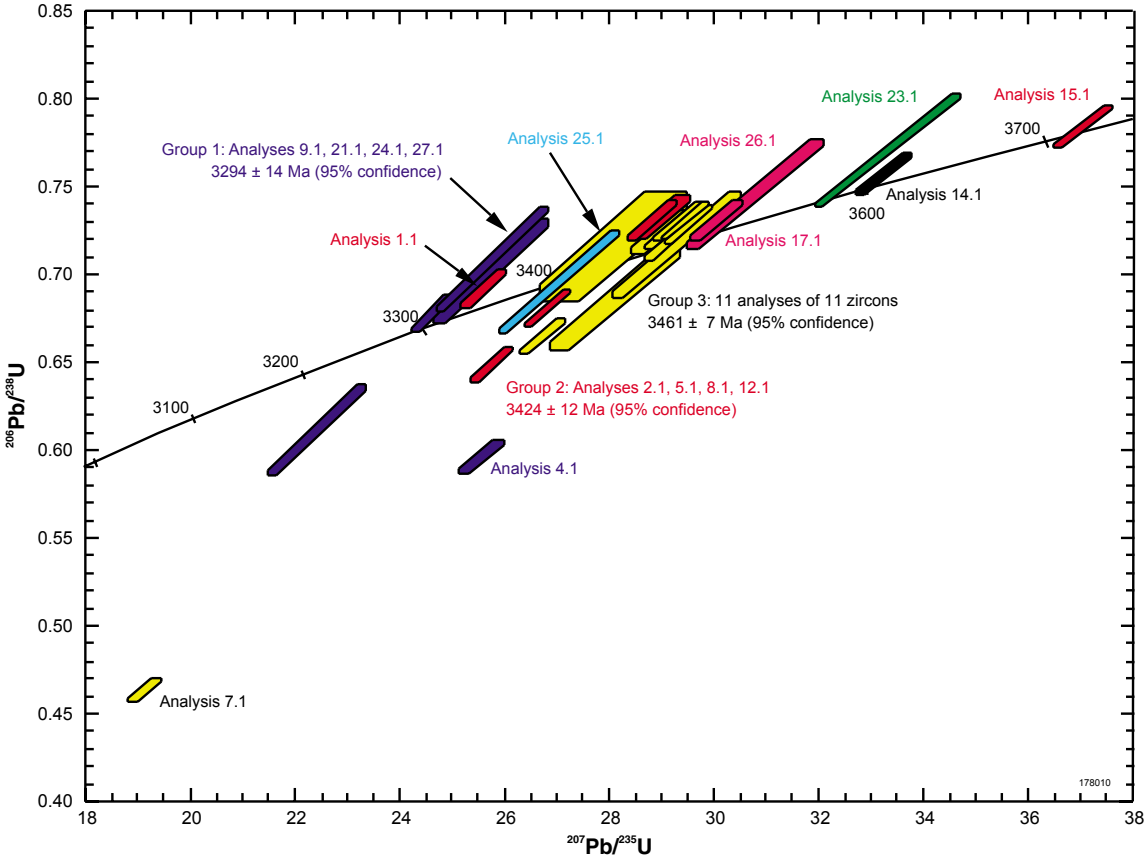

Figure 2. Concordia plot for sample 178010: lithic–quartz sandstone, Mount Olive

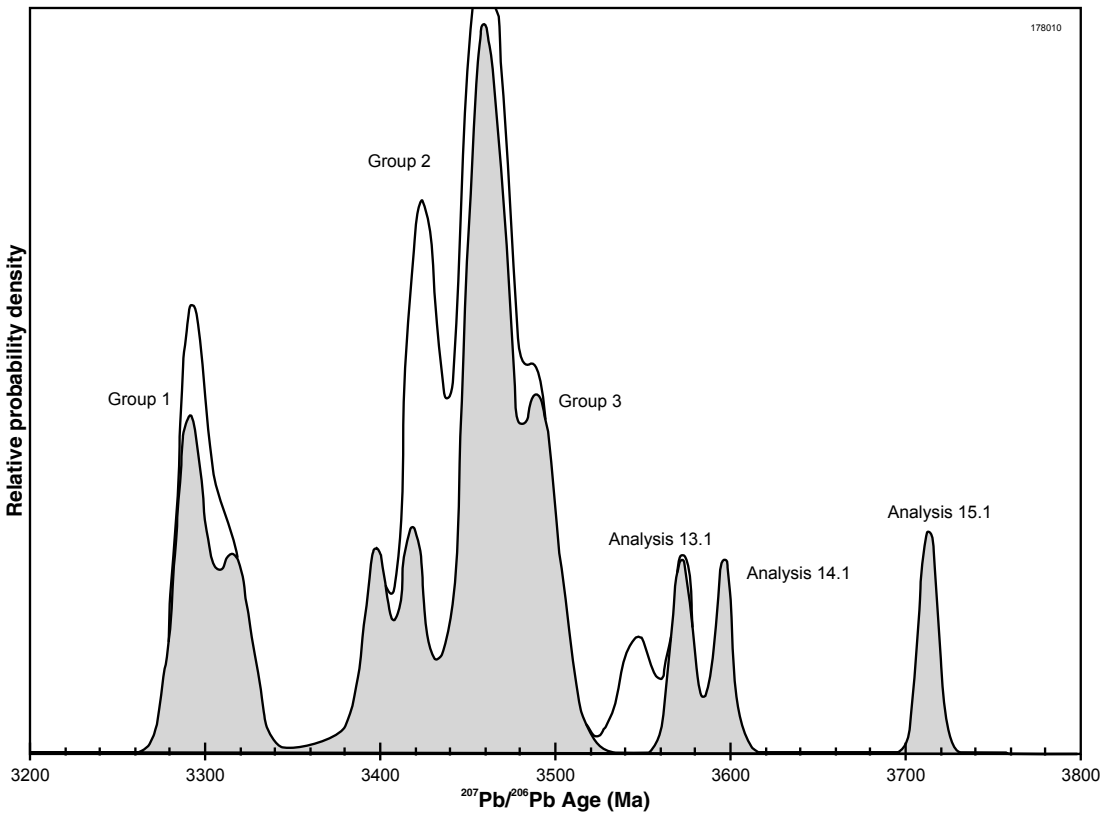

Figure 3. Gaussian-summation probability density plot for sample 178010: lithic–quartz sandstone, Mount Olive

Analyses 9.1, 21.1, 24.1, and 27.1 were obtained from small (generally less than 80 × 100 μm in size), euhedral grains and irregularly shaped grain fragments that show clear evidence of surface pitting and abrasion, consistent with an episode of detrital transport. The date of 3294 ± 14 Ma indicated by the weighted mean $^{207}Pb/^{206}Pb$ ratio of concordant analyses 9.1, 21.1, and 27.1, and slightly discordant analysis 24.1 of Group 1, is therefore interpreted to provide a maximum age for deposition of the sandstone. The older dates provided by the remaining analyses are also interpreted to be of detrital zircons. Possible source rocks within the western part of Australia having ages matching those of the zircons within this sample include rocks from East Pilbara Granite–Greenstone Terrane and the Narryer Terrane of the northwest Yilgarn Craton (Geological Survey of Western Australia, 2004).

Recommended reference for this publication:

NELSON, D. R., 2005, 178010: lithic–quartz sandstone, Mount Olive; Geochronology dataset 549; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

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